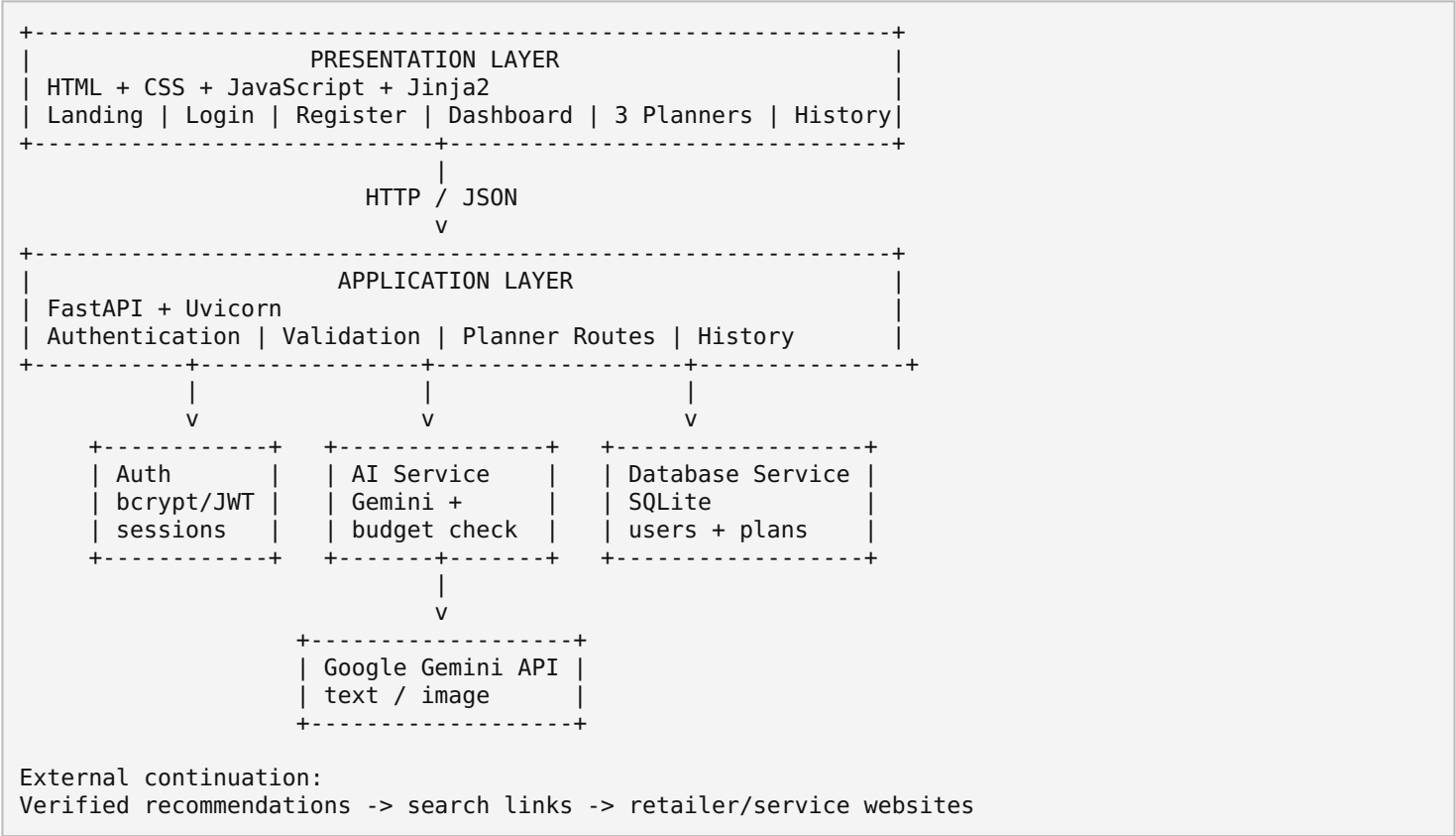


Solution Architecture

Date	03 October 2026
Team ID	SWTID-2026-5999
Project Name	PocketSmart AI — AI-Assisted Goal-Based Budget Planner
Maximum Marks	5 Marks

Architecture Diagram



Component Description

Component	Responsibility	Technology
Presentation	Collects inputs and displays plans, totals and history	HTML/CSS/JS/Jinja2
API Layer	Routes requests, handles sessions, validation and responses	FastAPI/Uvicorn
AI Service	Builds prompts, calls Gemini, parses structured output and supplies fallback data	Python + google-genai
Budget Engine	Recomputes category totals and keeps the result within the submitted budget	Python
Authentication	Password hashing, signed tokens and protected routes	bcrypt + JWT
Database	Stores users and generated recommendations	SQLite
Image Handling	Validates optional outfit uploads	Pillow
External Shopping	Creates search URLs rather than claiming live product availability	URL templates

Data Responsibility

The AI is responsible for **recommendation generation**. The application is responsible for **validation, calculation, access control and persistence**.